



Student **Management** System

Java-Based Academic Administration Project

SUBMITTED BY

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SUBMITTED TO

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Course: Object Oriented Programming (OOP)

Project Mission

A robust educational utility designed to handle complex institutional tasks including attendance tracking, grading logic, scholarship evaluation, and financial management using core OOP principles.

System **Core** Modules



Personnel

Advanced management of Student and Teacher profiles with unique ID tracking.



Finance

Automated fee processing with built-in validation to prevent duplicate payments.



Academics

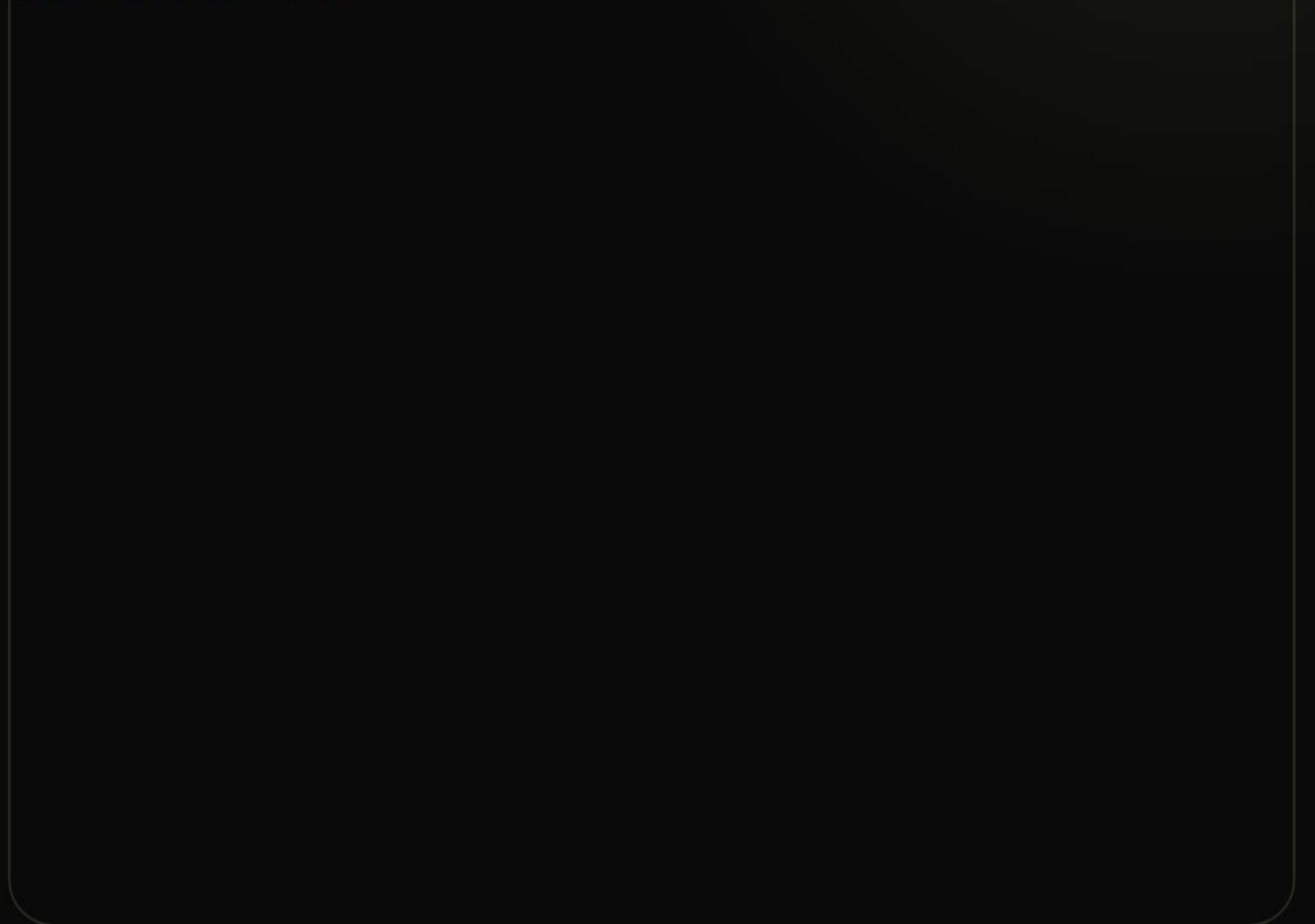
Real-time grading based on marks and automated scholarship eligibility.

Architectural Blueprint

The system architecture relies on a highly modular structure as shown in the **UML Diagram**:

- ✓ **Person (Abstract)**: Base class for all users.
- ✓ **Generalization**: Student and Teacher inherit from Person.
- ✓ **Aggregation**: Course holds a Teacher object.

 UML Class Diagram



OOP Hierarchy Design

Person

Attributes:

ID, Name, Email

Status:

Abstract Class

Student

Extended:

Semester, GPA

Method:

displayInformation()

Teacher

Extended:

Subject, Salary

Method:

displayInformation()

Secure Authentication

The system is protected by a login wall requiring a hardcoded admin username and password (admin/1234).

```
if(user.equals(username) && pass.equals(password))  
{  
    System.out.println("Login Successful!");  
}
```

Grading Logic Implementation

Criteria (Marks)	Grade	Status
≥ 90	A+	Outstanding
≥ 80	A	Excellent
≥ 70	B	Very Good
≥ 50	C	Satisfactory
< 50	Fail	Incomplete

Scholarship Eligibility

100%

CGPA $\geq$ 3.85

50%

CGPA $\geq$ 3.75

25%

CGPA $\geq$ 3.50

Automated calculation ensures fair and merit-based scholarship distribution.

Finance & Notifications

 Finance and Payment

Secure Billing

The **Fee Class** tracks payment status using boolean flags, ensuring each student only pays once per semester.

Messaging Service

The **Notification Class** centralizes all communication, allowing the admin to broadcast messages to the entire system console.

User Interface & Control



Console Interaction

Driven by a robust Do-While loop that keeps the application alive until the user chooses to exit (Choice 0).



Switch-Case Logic

11 different execution paths for viewing info, paying fees, checking scholarship, and manual data entry.



Data Capture

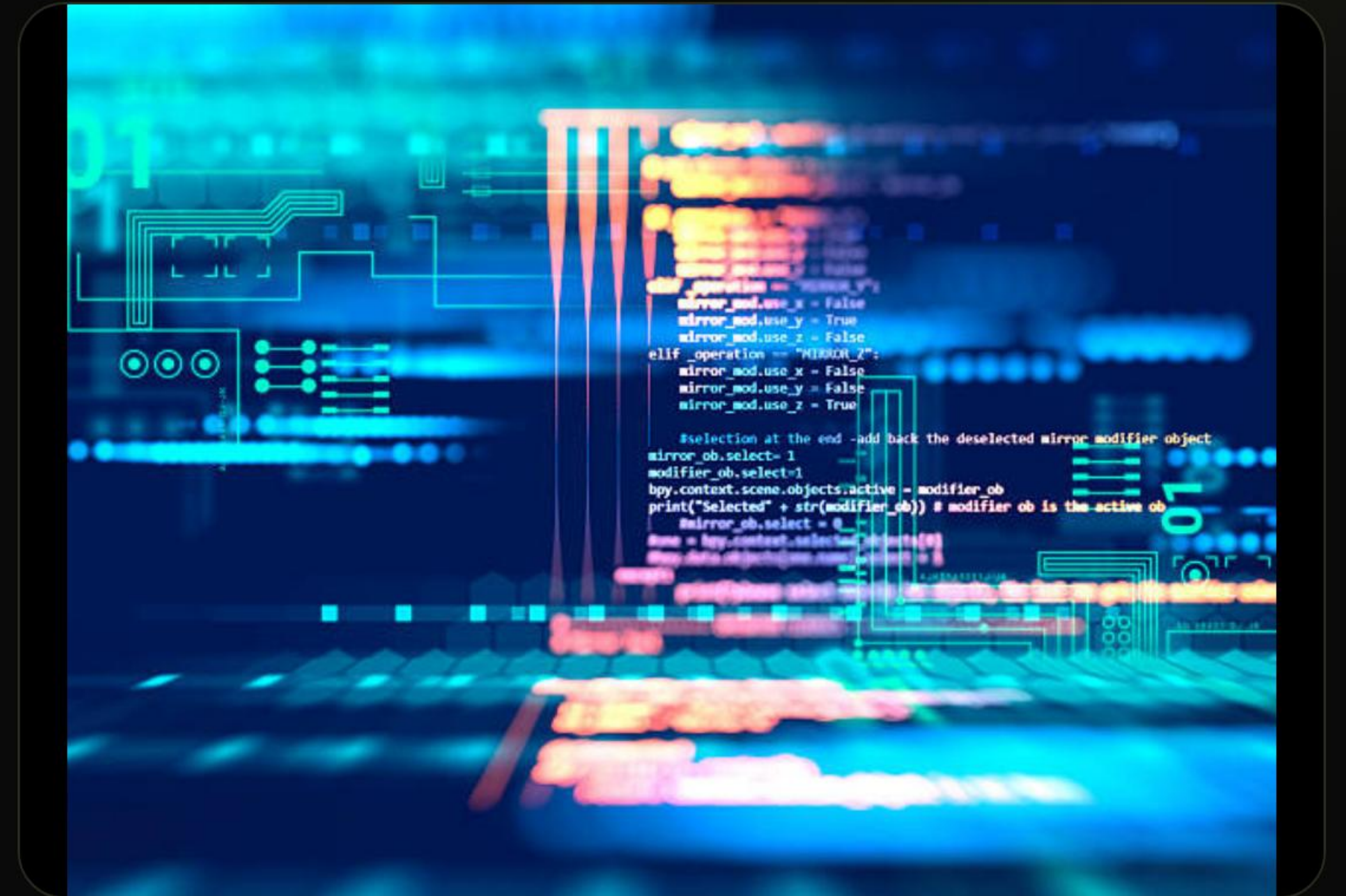
Uses Java Scanner to allow real-time manual updates for student and teacher records during runtime.

OOP Implementation Summary

Encapsulation: All attributes are private, accessed via protected getters.

Inheritance: Shared behavior in Person is extended to sub-classes.

Polymorphism: Method overriding in displayInformation().



Q&A

Thank You for Your Attention!

Muhammad Salman & Shaheryar Khan

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Image Sources



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